REPORT

OF THE

INSPECTORS OF IRISH FISHERIES

ON THE

SEA AND INLAND FISHERIES OF IRELAND, FOR 1890.

TO HIS EXCELLENCY LAWRENCE, EARL OF ZETLAND,

&c., &c., &c.,

LORD LIEUTENANT-GENERAL AND GENERAL GOVERNOR OF IRELAND

MAY IT PLEASE YOUR EXCELLENCY,

We have the honour, in conformity with section 112 of the 5 and 6 Vic., cap. 106, to submit our Report for the year 1890, being the twenty-second since the Sea and Inland Fisheries of Ireland have been placed under the superintendence of this Department.

THE SEA FISHERIES.

The returns obtained from the Collectors of Customs and the Coastguard show that the number of Registered Vessels in Ireland during 1890, fishing for sale, amounted to 5,655, with crews consisting of 21,122 men and 865 boys.

If these returns be correct, they show a still further decrease during the past year in the number of vessels and boats engaged in the fisheries of 112; while the number of men employed has increased by 547.

An analysis of the returns around the whole coast shows that the decrease mentioned is made up by a reduction of 15 first class and 135 second class boats, while there has been an increase of 38 third class boats.

It appears from them that on the east coast, from Howth, in the County Dublin, to Greystones, in the County Wicklow—a line of coast which embraces the harbours of Howth, Dublin (where there is a large number of some of the finest fishing vessels afloat), Kingstown, Dalkey, Bray, and Greystones—there is a *decrease* of 23 vessels—8 being first class, 12 second, and 3 third class.

Proceeding southwards, we find in the counties of Wicklow and Wexford, as far as Bannow Bay, a considerable *increase*, viz., 44 vessels, made up of an increase of 2 first class and 37 second class, and a decrease of 11 third class at Arklow, and an increase of 1 first, 10 second, and 5 third class in Wexford.

In Waterford there is a slight *decrease* of 5 second class boats only partially engaged at fishing. This brings us to a part of the coast east of Dungarvan, and from that, round the whole of the south, south-west, and west coasts, up to and including the Galway division, which extends to a point north of Kilkerrin Bay (with a few slight exceptions), the returns show a *general decline* in the number of craft engaged in fishing, amounting to no less than 116 boats on this stretch of coast. The exceptions are an *increase* of 14 second and third class boats about Youghal and Helvick Head, 6 second and third class at Castletown Bere, and 16 of same classes at Dingle.

The greatest decline on this part of the coast appears to be in the Queenstown Division, which embraces the finest harbour in Ireland, where there has been a reduction of 38 craft, being 1 first, 14 second, and 23 third class.

In the Skibbereen Division, which includes the fine harbours of Glandore, Castlehaven, Baltimore, Skull, Crookhaven, Long Island, &c. (some of the principal centres of the great mackerel fishery), there appears to be a reduction of 36 craft, second and third class. There is, however, an increase of 2 first class vessels "solely" engaged in fishing, and 2 of the same class "partially" so.

From Kilkerrin Bay, County Galway, to the north point of Clew Bay, near Achilbeg, county Mayo, which embraces a great portion of the Connemara coast, we find an *increase* of 31 third class boats.

At Achill a *decrease* of 6 craft, 5 being third and 1 first class.

Thence round the west and north-west coasts as far as Donegal, which include the fine natural harbours of Blacksod, Broadhaven, Killala, and Sligo, a *decrease* (save in one small place on the Sligo coast where there has been an *increase* of 3 third class boats) of 24 has taken place.

Between Donegal and Bloody Foreland, embracing Arran and the many islands on this part of the coast, there has been an *increase* of 3 first, 4 second, and 41 third class vessels.

From Bloody Foreland round the north, north-east, and east coast as far as Cranfield Point, Co. Down, we find a *decrease* of no fewer than 5 first and 70 second class, and an *increase* of 4 third class; making a total *decrease* in this part of the coast of 71 craft.

From Cranfield Point, Co. Down, to Howth, there has been a *decrease* of 2 first and an *increase* of 12 second and third class, making an *increase* of 10 boats.

In several Reports we have expressed our opinion that these returns cannot be relied on. From the details we have given, it is almost manifest that there are many inaccuracies.

The system of keeping the registry is, we believe, imperfect. There are many miles of coast unguarded, and from which it is difficult, if not impossible, to obtain returns, or enforce the registry of boats engaged in fishing.

A revision of the registry of all boats around the entire coast, and the adoption of an improved system of registration, is much required. We have, however, no funds at our disposal for the purpose.

Although we are of opinion that these statistics are not very reliable, we think it well to give, in a tabular form, the variations on the different parts of the coast we have described between the returns of 1890 and those of 1889:—

—	Increase.			Decrease.			Total Increase.	Total Decrease.
	1st Class.	2nd Class.	3rd Class.	1st Class.	2nd Class.	3rd Class.		
Howth to Greystones,	–	–	5	8	12	8	–	23
Greystones to Bannow Bay, co. Wexford,	3	47	5	–	–	11	44	–
Bannow Bay to Stradbally, co. Waterford,	–	23	8	–	31	5	–	5
Stradbally to Kilkerrin Bay, co. Galway,	14	66	212	19	161	228	–	116
Kilkerrin Bay to Achillbeg,	–	–	76	–	8	37	31	–
Achill,	–	–	–	1	–	5	–	6
Achill to Donegal,	–	–	11	–	4	31	–	24
Donegal to Bloody Foreland,	3	4	41	–	–	–	48	–
Bloody Foreland to Cranfield Point,	10	99	25	15	169	21	–	71
Cranfield Point to Howth,	1	12	5	3	1	4	10	–

As it may be considered desirable that these returns should be given more in detail without embracing such long stretches of the coast as we have given, we append the following return in Coastguard divisions :—

Division.	Extent of Coast.	Increase.			Decrease.			Total Increase.	Total Decrease
		1st Class.	2nd Class.	3rd Class.	1st Class.	2nd Class.	3rd Class.		
Kingstown,	Howth to Breeches, S. of Greystones,	–	–	5	8	12	8	–	23
Arklow,	Breeches to S. of Cahore,	2	37	–	–	–	11	28	–
Wexford,	Cahore to Bannow,	1	10	5	–	–	–	16	–
Waterford,	Bannow to near Stradbally,	–	23	8	–	31	5	–	5
Youghal,	Stradbally to Garryvoe,	–	20	25	3	18	13	11	–
Queenstown,	Garryvoe to Minane River,	7	–	–	8	14	23	–	38
Kinsale,	Minane River to Galley Head,	2	17	49	4	22	48	–	6
Skibbereen,	Galley Head to Snave Bridge,	4	1	4	–	22	23	–	36
Castletown,	Snave to Kenmare,	–	1	5	–	–	–	6	–
Valencia,	Kenmare to Castlemaine,	1	–	14	2	10	13	–	10
Dingle,	Inch Point to Blennerville,	–	1	48	–	11	22	16	–
Ballyheige,	St. Annagh to Foynes,	–	6	15	1	6	16	–	2
Seafield,	Ballymacrana to Cancapple,	–	1	42	–	1	69	–	27
Galway,	Derreen to Mace Head,	–	19	10	1	57	1	–	30
Clifden,	Mason Island to Doaghbeg,	–	–	76	–	8	37	31	–
Keel, Achill,	Doaghbeg to Fahy,	–	–	–	1	–	5	–	6
Belmullet,	Fahy to Brandy Point,	–	–	–	–	3	6	–	9
Ballycastle, Mayo,	Brandy Point to Bartra,	–	–	4	–	–	18	–	14
Pullendiva,	Ballina to Coney Island,	–	–	4	–	1	–	3	–
Sligo,	Coney to Donegal,	–	–	3	–	–	7	–	4
Killybegs,	Donegal to Gweebarra,	4	2	8	–	–	4	10	–
Guidore,	Gweebarra to Bloody Foreland,	–	5	37	1	3	–	38	–
Rathmullen,	Bloody Foreland to Dunaff,	1	–	1	1	15	8	–	22
Moville,	Dunaff Head to Culmore,	1	–	–	–	14	–	–	13
Ballycastle, Antrim	Down Hill to Garron,	1	25	4	–	43	–	–	13
Carrickfergus,	Garron to near Belfast,	–	3	10	–	–	11	2	–
Donaghadee,	Belfast to S. of Cloghy,	–	41	2	4	33	3	3	–
Strangford,	Cloghy to Sheepland Head,	–	–	1	1	29	2	–	31
Newcastle,	Sheepland to Derryogue Point,	7	30	7	9	24	8	3	–
Dundalk,	Derryogue to Maiden Tower,	1	8	4	2	–	–	11	–
Malahide,	Maiden Tower to Howth,	–	4	1	1	1	4	–	1

These details will give some idea of the increase and decrease, as stated in the returns, on the several portions of the entire coast.

In Appendix No. 1, p. 102, will be found a return of the number of boats and men engaged in fishing in the several divisions named above, and whether solely or partially so engaged.

Seasons for Sea Fish.

In Appendix No. 22, p. 150, will be found a Table showing the Seasons during which the principal descriptions of Sea Fish are taken round the Coast of Ireland.

Investigations on the West Coast.

During the past year the Royal Dublin Society, aided, as we are informed, by the Irish Government, undertook a series of experiments on the west coast. They were confided to the charge of our colleague, Mr. Green, whose report we append. It exemplifies more fully than anything we can say the importance of having a vessel permanently attached to this department, a matter which we have for several years felt it our duty to bring under the notice of the Government.

Mr. Green reports as follows :—

"So long ago as 1821, we find the Commissioners of Irish Fisheries reporting to Government as follows :—

"'The Commissioners conceive it will not be irrelevant to the subject which constitutes the immediate part
"'of this report, to introduce an extract from the Report of a Committee of the Royal Dublin Society, to whom
"'were referred the statistical surveys of this country, together with Mr. Frazer's very able publication on its
"'fisheries. 'Your Committee entertain, in common with the Society, the most sanguine hopes that great
"'advantages will result from the useful undertaking of the Board of Fisheries in having surveys, and plans,
"'and estimates made at all the fishing stations along the whole coast of Ireland. Your Committee is of
"'opinion, that with a view of selecting plans for commencing new harbours it would be an object of the
"'greatest importance that the Royal Dublin Society should aid the exertions of the Board of Fisheries, by a
"'statistical survey of the coast.'"

"During the past year it fell to my lot to act in conjunction with the Royal Dublin Society in a manner somewhat in accord with the above suggestion made seventy years ago. The Government and the Society sharing the expenses of the work.

"The object aimed at was to test the value of the fishing grounds off the west coast of Ireland experimentally and report on them. This seemed all the more necessary after studying the already published reports of such fishing grounds.* In them we find voluminous evidence purporting to give information on the subject, but when one comes to study the location of fishing grounds and their present condition, the evidence is so contradictory, so biassed by prejudice, so much of it is given by persons quite ignorant of the coast beyond their own particular bay, and by men incapable of describing what probably in a practical way they know very well, that it is almost impossible to arrive at any reliable conclusion. As illustrative of this difficulty the following quotation from the Commission of 1835, compared with our experience will suffice—

Report of 1835.

"'A great bank . . . is ten leagues westward
"'of Slyne Head and Achil Head . . . the sound-
"'ings are from thirty to sixty fathoms . . . great
"'quantities of fish have been taken on it by wherries of
"'from twenty to forty tons burden, belonging to
"'Skerries on the Dublin coast.'"

Extract from our Log in 1890.

"'At 12 o'clock, noon, when about thirty miles (*i.e.*
"'ten leagues) westward of Achill Head, the engines
"'were stopped and a sounding made in 220 fathoms,
"'bottom fine sand; surface temperature 56° F.;
"'bottom temperature 50° F. We immediately shot the
"'eighteen-foot beam trawl and veered to it 550 fathoms
"'of warp. The fish captured were all of deep sea forms
"'(*i.e.* fish quite unknown in the markets.)'"

"Here it will be seen that on the spot described as a bank with thirty to sixty fathoms there is in reality a depth nearly four times as great. I have quoted from this old report because it is the account of one of the most minute and careful investigations ever undertaken in the case of the Irish Fisheries, and as it describes the state of the fisheries before the period of the great famine, and when many boats helped by the fishing bounties then in vogue, flocked from all parts to the west coast and fished in places where now a line is never cast, it contains a mass of information of great value, but useless unless some clue be found to guide to the truth. That clue can in no other way be found except by actual experiment.

"The Royal Dublin Society, therefore, having authorized me to make such experiments, I proceeded to organize an expedition for the exploration of the fishing grounds of the West Coast of Ireland, with a view to ascertain (1) their extent, (2) their productive value, (3) how far they are being utilized, (4) in what direction they can best be developed for the benefit of the coast population.

"Their extent could best be made out by reference to the Admiralty charts, by practical experiment, and from the knowledge possessed of them by local fishermen. Their productive value by experiment, by the statistics collected by the Board of Irish Fisheries, by local information, and by scientific investigation concerning the reproduction, migration, and food supply of fish.

"As the work before me covered a very wide field, it was most desirable that I should have helpers skilled in the various points I proposed to investigate. I, therefore, considered myself fortunate in being able to secure the aid of Professor A. C. Haddon, of the Royal College of Science, who had been my companion on two former expeditions for marine research. For the topographical work I engaged Mr. T. H. Poole, C.E., an enthusiastic yachtsman and navigator, who had also been with me on two deep-sea expeditions, though not the same as those on which Professor Haddon had been my companion. He was to be second in command of the expedition, and to carry on the work in case of my being absent on other duties. Owing to collegiate engagements, Professor Haddon could not join the expedition till the end of June; and as I was anxious to commence work on the 1st of May, we succeeded in finding a worthy substitute in Professor Prince, of Glasgow, well known as being joint author with Professor MacIntosh of a fine work on the development of Teleostean fishes. Dr. Scharff, Curator of the

* Report of the Commissioners of Inquiry into the state of the Irish Fisheries, 1836. Appendix II., 7.

Science and Art Museum, Dublin, also rendered valuable help for the first portion of May, and when Professor Prince left us early in June, Mr. Ernest Holt took his place. Besides knowing Mr. Holt from his publications on Fishes, I had met him at St. Andrews, where I went, shortly before starting on our cruise, to see something of the researches being carried on there under Professor M'Intosh's guidance. Mr. Holt remained with us for nearly two months, and his valuable paper accompanying this report must speak for the work he accomplished. Mr. W. de Vismes Kane helped us when Dr. Scharff left, and was followed by Mr. Duerdon, of the Royal College of Science, and later on by Mr. Francis Dixon, B.A., T.C.D., who stayed with us till the end, and rendered much valuable assistance. Mr. R. Ussher also joined us for a short time to study the Bird fauna of the West coast.

"No little difficulty was experienced in finding a suitable steamer for our work. A great many steam yachts with ample cabin accommodation were available, but were deficient in the requirements for working our fishing gear. Steam trawlers were available, but without any sleeping accommodation beyond what was barely necessary for the crew; and for a short charter of four months, at most, it would not pay owners to make those alterations in their vessels that we considered necessary for our accommodation and our experiments.

"In the end, I was fortunate in finding the steam yacht *Fingal*, which the owner had been using as a fishing vessel in the North Sea; and, with slight alterations which the owner agreed to, she was fitted for our purpose. She was 110 feet long, 17 feet 9 beam, and about 200 horse power, effective, and fully rigged as a schooner yacht.

"Our crew consisted of:—

Skipper,	JOHN GREEN,	}
Engineer,	WALTER RUGMAN,	} of Sunderland.
Fireman,	WILLIAM RUGMAN,	}
Mate,	WILLIAM POWER, Ringsend.	
Seaman,	CHARLES OLDEN, Cork Harbour.	
Do.,	JEREMIAH DRISCOLL, Courtmacsherry, Co. Cork.	
Do.,	JAMES MORGAN, Aberdeen.	
Steward,	WILLIAM WILSON, Cork Harbour.	
Boy,	ARTHUR GODFREY, do.	

"About half of our crew were accustomed to fish in the North Sea, while the rest with the exception of the steward and boy, were experienced Irish fishermen.

"Our work comprised trawling, long-line fishing from the steamer, trawling with our nets from local fishing boats, and a few shots of our mackerel nets from an Arran Island hooker. Our trawling experiments extended in some places from a depth of 500 fathoms, and fifty miles from land to the shallow bays, while the long lines were shot in depths varying from 100 to 20 fathoms.

"To procure a train of mackerel nets which would enable us to experiment fully on the existence of schools of spring mackerel where the fishing is not at present prosecuted, seemed too expensive an item for our outfit, so the experiments made with the short train of six nets were quite too limited to afford reliable evidence, when that evidence was almost negative, for any important generalization.

"The following is a list of fishing gear, scientific apparatus, &c., which formed our equipment:—

One-inch steel wire trawl warp, 160 fathoms long.
$\frac{7}{8}$-inch " " 750 "
$\frac{3}{4}$-inch " " 500 "
Beam trawl, ordinary pattern, 25 ft. beam.
" patent discriminating trawl, 25 ft. beam.
" lined with fine netting for deep sea, 18 ft. beam.
One otter trawl.
One 9-foot Agassiz double trawl, lined with mosquito net.
Shrimp trawl for catching fish fry, 9 ft. beam.
Dredges of various types.
A train of six mackerel nets and warps.
Surface nets for fish ova.
Six baskets of long lines, about two miles in all, with buoys, anchors, &c.
Steel wire sounding machine, with 1,400 fathoms of wire.
Sieves for sorting, and tanks of methylated spirits for preserving specimens.
Bulbs for testing specific gravity.
Deep sea and surface thermometers.
Discs for testing transparency of water.
Two microscopes.
A good equipment of laboratory appliances and reagents.
A specially selected library of books on Marine Biology.
Books of schedules for recording observation, viz.:—
No. 1.—Log book.
No. 2.—Trawling record.
No. 3.—Long-line record.
No. 4.—Condition of reproductive organs and stomach in fish caught.

"Some of the gear had been used by me on former expeditions when working in depths ranging to 1,000 fathoms, but the greater part of it was new. The sieves, tanks, thermometers, microscopes, and laboratory apparatus belonged to Professor Haddon, and were kindly lent by him.

"The *Fingal's* long-lines being thirteen miles long, only a portion of them was taken, my intention being not to aim so much at capturing great quantities of fish as to test the grounds sufficiently. This also was our reason for not taking larger trawls.

"The data as to the physical conditions under which the fish captured, and of their number and condition, are recorded in the books of schedules, all of which were carefully kept, and were arranged as follows:—

"No. 1.—The Log Book contains a record of position, date, hour, depth, nature of bottom, barometer, thermometer (air, surface, and bottom), specific gravity of water state of tide, wind, and weather of every haul of the trawl, dredge, or shot of long lines.

"No. 2 records the locality, depth, nature of bottom, trawl used, time occupied in trawling, the gross number of fish taken, their average size and weight, proportion of males to females, and the bottom and surface fauna.

"No. 3 gives a similar record for the long-line fishing.

"No. 4 gives some precise statistics as to size and weight of some of the fish taken on a given day. Some 2,000 fish were treated in this complete manner, besides very many more which were more cursorily examined.

"We were most unfortunate in the weather which prevailed throughout our whole cruise. For the first half of May there was a promise of summer, but then a gale set in, and from that until September we experienced little more than a series of storms, accompanied by dark mists and drizzling rain. There were occasional fine days, but the fine weather never lasted for a week, so our deep-sea work was much curtailed. Once or twice we hauled our lines in a sea which was running so high as to make it utterly useless to attempt trawling. The presence of enormous schools of dog fish on the coast this summer was also most unfavourable for line-fishing, and from accounts received from the west of Scotland, it seems that these pests of the ocean injured the fisheries for a long way to the northward.

"Our cruise commenced on the 8th of May at Queenstown, and lasted till August 30. During that time we examined the Kenmare River, Ballinskelligs, Dingle, Smerwick and Brandon Bays; and fished to depths of 80 to 100 fathoms off the Skelligs and Blaskets on the Kerry coast. Paying only flying visits to Carrigaholt and Liscannor, on the coast of Clare, we made the Arran Islands our headquarters while exploring the fishing grounds of Galway Bay.

"This fine bay would require a much more lengthened and minute study than we could afford to give to it. Here, in the years 1873 to 1876, our Board organized a series of very important observations, coastguards were placed on board the trawlers, and throughout a period extending over three years, and comprising each month of these years, the fish captured were carefully recorded, and the position of each haul marked on the chart. From these records, the migrations of fish in the bay may be studied. The chief object of these experiments was to test the accuracy of the charge that the trawlers were destroying fish spawn, and an undue amount of small fish. To these questions, the conclusions arrived at afford a complete negative, which is not to be wondered at when the isolated and pelagic nature of fish eggs is taken into account. The weakness of these experiments lay in the fact that the observers though very careful men were unskilled in the main aspects of the case they had to deal with. Galway Bay has been the scene of the most violent struggles between the various classes of fishermen, and we think many of the burning questions which led to these disputes are still unsolved by science. That they are possible of solution, I have no doubt, but nothing but careful and oft repeated observations will form a basis on which any reliable solution can be founded. On leaving Galway Bay, and having explored the large bays of South Connemara, we fished to the north-westward of Slyne Head, also on the banks off Inishboffin, and Clare Island in Clew Bay. Blacksod Bay and Broad Haven; and off Achill our experiments were carried to a distance of fifty miles from land into the great ocean depths.

"Our most northern port was Killybegs, in Donegal, which we made our headquarters for more than a week while we experimented in Inver Bay which has been closed against trawling for forty years.

"We then worked our way southwards, and gave up the *Fingal* in Dublin on the 31st of August.

"Several reports on our work are in preparation. Mr. Holt's paper on fish development—"The Eggs and Larvæ of Teleosteans"—is the first of a series, which in due course will appear, on special subjects which we investigated.

"In this paper, which is illustrated by six carefully drawn coloured plates, and a diagram to scale, showing the relative size of fish ova, he describes the capture of the eggs or larvæ of twenty-nine species of fish, no less than nine of which have now been recognised for the first time. Many of these larval fish were hatched out of the eggs in small glass pans of sea water on board.

"Although a report on our experiments with the patent discriminating trawl has already been laid before my colleagues and incorporated in the report of our Board on the subject, to his Excellency the Lord Lieutenant, it may not be out of place to repeat the general results arrived at.

"The patent discriminating trawl aims at being a net which will catch fish as well as the ordinary beam trawl, but, at the same time, will not crush the fish to death while the haul is being made, and will also admit of small immature fish escaping uninjured.

"The patentee hopes that his net may, therefore, be of importance, whenever occasion arises to prohibit or limit the use of the ordinary beam trawl.

"The patent trawl differs from that in ordinary use by the bag not tapering to a narrow cod end, but being square all the way down, so that the end of the bag is exactly the same breadth as the beam. This gives ample room for the fish to swim about and not get crushed or sanded up. The mesh is also peculiar. Instead of the threads running diagonally, the meshes are set up square to the beam, like a tennis net. This arrangement is for the purpose of preventing the meshes closing up when strain is created by the weight of fish or weeds in the bag. Small fish have, therefore, no hindrance in leaving the net, provided the mesh is large enough for them to pass through. But it is obvious that there is an extra strain on every second thread—that is, on all those running parallel with the sides. Those crossing these at right angles can take none of the weight. The twine must, therefore, be twice as strong as the common trawl. This made the net cumbersome and heavy compared with the ordinary one, but it killed fish as well, was not so liable to collect sand, and no doubt a number of small fish escaped through the meshes. However, when sole are fished for, it is impossible but that a mesh small enough to retain them uninjured must also retain a number of unmarketable plaice, brill, turbot, &c. Fry of two or three inches in length have a much better chance of escape in the patent trawl. There are some other details in the get up of the net, but as the funnel included in the patent has been in use for over twenty-five years on various parts of the coast, the above are the special features worthy of attention.

"Every haul we made and every shot of our lines has been carefully marked on our charts, but as we are now about to start on a new season's work, I reserve any general conclusions till our survey is complete.

"I have met with much greater difficulty in finding a steamer for the work in 1891 than in 1890. The *Fingal* is not available, having gone away to the Mediterranean, and the *Harlequin* which, after three months correspondence I have fixed on, is so much more expensive a vessel, that in order to keep expenses within the funds available, it has been necessary to limit our cruise to one of three months. As it is important that the season of our experiments should vary, and as the summer months were selected last year, I have now determined to test the grounds in March, April, and May. The probability of encountering storms in spring, is, of course, greater than in summer. But this risk is unavoidable if the deep-water grounds are to be tested when they are most likely to be productive. We tried, as much as possible, to let the coast fishermen have an insight into our work. Numbers of them came on board the *Fingal*, and by taking our gear in the local fishing boats, I was able, in some instances, to suggest new modes of fishing which have in certain places been taken up and acted on with success. Owing to our time being limited, there were many details passed over, and many a bay with its flotilla of canoes and its fishing hamlets had to be looked at only in passing. Should the weather prove more favourable in our three months cruise in 1891, than it did in the past year, many blanks may be filled up. Should, however, it prove adverse, we must look forward to leaving many things untouched that stand in need of careful investigation."

This report demonstrates the difficulties under which we have laboured, in having been precluded from the possibility of making important experiments. The services of a suitable steamer have been asked for during several years past. Chartering a steamer for a limited time does not meet the necessities of the case. The months fixed on in the charter may, on account of weather or other causes, prove the worst in the year. A considerable outlay is lost each time in providing gear to suit the vessel, and no vessel hired for a short term can be equipped in a manner thoroughly complete. Money has been lost on fishery loans issued to the island populations, chiefly owing to the impression,

that, in consequence of the difficulties of access, responsibilities might be repudiated with greater safety than on the mainland. This is demoralizing and unfortunate, because it affects those places much in need of the benefits of the loan funds. Lastly, it is chiefly from sea, and not from land, or by correspondence, that the real wants of the coast fishing population can best be studied, the true state of the case arrived at, and practical instructions given when occasion arises. We beg particularly to refer to our reports on this subject for the years 1871, pp. 7 and 8; 1873, p. 8; 1874, p. 7; 1875, p. 6; 1876, p. 4; 1877, p. 3; 1878, p. 4; 1879, p. 4; 1880, p. 3; 1881, p. 4; 1882, p. 3; 1883, p. 3; and 1886, p. 8.

MACKEREL FISHERIES.

The numbers of vessels and boats of each nationality employed at the mackerel fishery of 1890 were as follows:—

English and Manx,	269
Scotch,	23
French,	71
Irish,	300

The numbers engaged in 1889 were:—

English and Manx,	288
Scotch,	68
French,	68
Irish,	288

Spring Mackerel Fishery.

The mackerel fishing in spring is mainly carried on off the south and south-west coast of Ireland. In other places mackerel fishing does not commence till late in the year, when it is also carried on off the south and south-west coasts; so that the latter have now the advantage of two seasons' fishing—spring and autumn. Nearly all the fish taken in spring is exported, *fresh*, packed in ice in boxes. Nearly all the autumn fish is now salted in barrels for the American market.

The spring fishing is carried on chiefly by first class fishing vessels belonging to England, Scotland, Isle of Man, France, and Ireland; and in addition, by a large number of small boats or curraghs belonging to some Irish stations. The former use large drifts nets; the latter drift nets and seines.

The quantity of mackerel taken in the spring of 1890 (twelve weeks' fishing) was 206,122 boxes; and the amount realized to the fishermen £173,687.

The particulars from each station, with the total amount realized, and the average prices, are given in the following schedule:—

Place.	Boxes.	Average Price per Box.	Total Value.
		£ s. d.	£
Castletownbere,	49,693	0 18 11	46,995
Baltimore,	40,567	0 16 8	33,914
Kinsale,	35,026	0 16 8	29,166
Ventry,	27,013	0 11 5	15,452
Crookhaven,	13,902	1 2 10	15,865
Valencia,	10,724	0 12 3	6,583
Castletownshend,	6,181	0 15 4	4,753
Dingle,	5,284	0 16 5	4,335
Brandon,	4,160	1 0 0	4,160
Union Hall,	3,362	0 9 5	1,585
Ballycrovane,	1,929	1 1 0	2,025
Ballydavid,	2,717	1 8 5	3,860
Schull,	1,610	0 12 0	961
Caherciveen,	1,556	1 3 10	1,858
Ballydonegan,	1,217	1 1 8	1,320
Kilkee,	300	0 11 4	170
Portmagee,	435	0 16 6	358
Kilcredane,	267	0 13 0	174
Ballinskelligs,	126	1 0 0	126
Seafield,	27	0 13 6	18
Liscannor,	16	0 5 0	4
Courtmacsherry,	10	0 10 0	5
	206,122	0 16 10	173,687